Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Lazo Christou	lazoc@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

For my final project, I wanted to do something that would help all the chaos that comes with traveling. So, I wanted to create a travel planning website that helps people organize their trips and stay within their budget. I enjoy traveling and planning trips, but I know it can be difficult to keep track of destinations, activities, transportation, and expenses all in one place. You want to enjoy your travel vacation, not stress. My goal is to create a website where users can add destinations they want to visit, plan activities for each day, enter estimated costs, and track their total spending. I also want users to be able to choose their budget and see whether their planned trip is within that amount. This website would make planning trips easier, more organized, and less stressful, especially for college students who want to travel without spending too much money.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

After research, there are several websites and apps that already help people plan and organize their trips. One example is **Wanderlog**, which allows users to create travel

itineraries, organize activities, set budgets, and track expenses. Another one I found is **TripIt**, which helps travelers keep their flight, hotel, and transportation reservations organized in one place. **Google Travel** also helps users research destinations, compare flights and hotels, and organize travel information. These websites have similar goals to my project, but can be a lot and get confusing. I want to focus on creating a simpler travel planner that is easy for college students to use, especially when planning trips on a budget.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

The purpose of my website is to make travel planning easier and more organized, especially for college students or anyone who wants to travel on a budget. I want users to be able to enter their destination, set a budget, add activities they want to do, and keep track of their estimated expenses. The website would ultimately calculate the total cost of the trip and let users know if they are staying within their budget or going over it. I also want users to be able to view their planned activities and expenses all in one place, rather than going to multiple websites or apps. Overall, my goal is to create a simple and useful website that makes planning a trip less stressful and helps people manage their money while traveling.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The goal of my program is to take user inputs about their travel plans, use calculations and conditional statements to organize expenses and compare costs to their budget, and show a summary of their trip to help them stay organized and manage their spending.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will enter their travel destination, total budget, number of days they plan to travel,

and estimated costs for transportation, accommodations, food, and activities. They'll also be able to add specific activities they want to do, along with the estimated cost of each activity.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program will calculate the total estimated cost of the trip by adding together the expenses for transportation, accommodations, food, and activities. Then it compares the total cost to the user's budget using if statements to determine whether they are over or under budget. The program will also use loops to organize and calculate the costs of multiple activities.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will receive an organized summary of their trip that includes their destination, planned activities, total estimated expenses, and remaining budget. The website will also display a message letting them know if they are within their budget or have gone over it. This will help users see where their money is going and make changes to their travel plans if needed.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

The main things my program needs to track are the Trip, Destination, Budget, Expenses, and Activities. Each trip will have a destination, a total budget, and a list of planned activities and expenses. Each expense will have a category and an estimated cost. These will help the program organize travel information, calculate total spending, and keep track of how

much money the user has left in their budget.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need: For each trip, I will need the destination, number of travel days, total budget, and estimated expenses. These details will help the program organize the trip, calculate the total cost, and determine if the user is staying within their budget.

Details I will ignore: I will ignore details such as the weather, specific flight numbers, hotel ratings, and transportation companies because they are not necessary for calculating expenses or organizing a basic travel plan.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

I think the most difficult part of this project will be creating the different pages and making sure they all work together. I also think keeping track of the user's inputs, organizing expenses into lists, and updating the budget calculations when new activities are added could be very hard. Another difficulty will be making sure the website runs correctly and doesn't crash when users enter unexpected information. There are a lot of different possibilities for answers, and it's a long code so I will have to check every single detail.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I plan to use AI to help me understand coding concepts, troubleshoot errors, and get guidance when I am stuck. However, I want to make sure I actually understand and apply what we have learned in class. I will do this by asking AI to explain how the code works, breaking it down into smaller steps, and connecting it to concepts we learned in class. I will also test the code myself, check for errors, and make sure any suggestions from AI work correctly with the Drafter library before adding them to my project.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My ultimate goal for this project is to create a travel planning website that is easy to use, organized, and visually appealing. I want to meet all the project requirements while also challenging myself to become more comfortable with Python. More importantly I want to attract buyers to my website. By doing this project I will naturally improve my understanding of functions, loops, dataclasses, and user inputs by applying what I have learned in class. I also want to create something that I could actually use when planning my own trips in the future.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to create a simple travel planning website for college students who want to travel on a budget. What are three common challenges people face when planning trips, and what are three useful features I could include to make my website easier to use and help people stay organized?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

Some of the main challenges people face when planning trips are staying within their budget, keeping track of different expenses, and organizing activities. Three useful features suggested were a budget tracker that calculates total expenses, an activity planner where users can add things they want to do, and a trip summary that displays all their plans in one place. Existing websites like Wanderlog, TripIt, and Google Travel offer similar planning tools. However, I want my website to be simpler and focus more on helping college students plan affordable trips without feeling overwhelmed.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I think creating a detailed activity planner that organizes everything into a daily schedule is too complicated for my skills. Since I am still learning Python and Drafter, I want to focus on something that I can realistically create and understand. Is there a simpler way to let users add activities and their estimated costs without having to create a full daily itinerary?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

After brainstorming, I realized that I should focus more on the budgeting side of travel planning instead of trying to include too many features. My website will mainly allow users to enter their destination, set a budget, add activities and expenses, and see how much money they have left. Instead of creating a complicated daily schedule, I can have a simple list of activities and their costs. This will make the project easier to code while still creating a useful website that meets the class requirements.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck, I will use class notes, lecture examples, and Drafter documentation to help me understand what I am doing wrong. I can also ask my professor or TA for help during office hours. I plan to use ChatGPT to explain confusing concepts, help me understand error messages, and suggest ways to fix problems. I will also look back at previous assignments to see how we used similar coding concepts in class.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

If I run into an error with my website, I will first read the error message carefully to figure out what went wrong. Next, I will check my code for common mistakes like spelling errors, missing parentheses, incorrect indentation, or wrong variable names. I will then test smaller sections of my code to narrow down where the problem is happening. If I still cannot figure it out, I will compare my code to class examples or ask ChatGPT to explain the error and suggest a solution. After making these changes, I will test the website again to make sure everything works correctly. If the issue continues, I will reach out to my TA or professor for help.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I put all my thoughts and ideas into ChatGPT to help brainstorm, organize, and develop my travel budgeting website idea for my final project. I will use it to explore possible features, organize my project, and better understand how I could meet the assignment requirements using Python and Drafter. Below is my full conversation with ChatGPT, including my questions and the AI-generated responses.

Conversation topic: Computer Science Final Project — Travel Planning and Budgeting

Website

Prompt 1 — Initial Brainstorming

I want to create a simple travel planning website for college students who want to travel on a budget. What are three common challenges people face when planning trips, and what are three useful features I could include to make my website easier to use and help people stay organized?

Prompt 2 — Narrowing the Project

I think creating a detailed activity planner that organizes everything into a daily schedule is too complicated for my skills. Since I am still learning Python and Drafter, I want to focus on something that I can realistically create and understand. Is there a simpler way to let users add activities and their estimated costs without having to create a full daily itinerary?

Conversation 1: Brainstorming My Project Idea

My Prompt:

I want to create a simple travel planning website for college students who want to travel on a budget. What are three common challenges people face when planning trips, and what are three useful features I could include to make my website easier to use and help people stay organized?

Summary of ChatGPT's Response:

ChatGPT identified three main challenges that people face when planning trips:

1. Staying within their travel budget and keeping track of expenses.
2. Organizing activities, destinations, and transportation.
3. Keeping all their travel information in one place.

ChatGPT discussed three possible features for my website:

1. A budget tracker that calculates estimated expenses and compares them to the user's total budget.
2. An activity planner that allows users to enter activities, destinations, and estimated costs.
3. A trip summary that displays the user's travel information and expenses in one organized place.

ChatGPT also discussed existing travel planning websites such as Wanderlog, TripIt, and Google Travel. These examples helped me understand what features already exist and how

I could make my project simpler for college students.

Conversation 2: Making My Project More Realistic

My Prompt:

I think creating a detailed activity planner that organizes everything into a daily schedule is too complicated for my skills. Since I am still learning Python and Drafter, I want to focus on something that I can realistically create and understand. Is there a simpler way to let users add activities and their estimated costs without having to create a full daily itinerary?

Summary of the Simpler Approach Discussed:

Instead of trying to create an entire daily itinerary, I could focus on making a basic activity list.

Users could enter the name of an activity and its estimated cost. The program would store these activities in a list and use a loop to calculate their total cost.

This would be easier to implement using Python concepts such as lists, loops, functions, and conditional statements.

It would also make my website more manageable for an introductory programming course.

Conversation 3: Input, Processing, and Output

Topic Discussed:

How my travel budgeting website could use the Input-Process-Output model.

Summary of the Project Planning:

Input: Users would enter their travel destination, number of days, total budget, transportation expenses, hotel costs, food expenses, and activity costs.

Processing: The program would add the estimated expenses together and compare the total cost to the user's budget using conditional statements. Loops could help calculate expenses from multiple activities.

Output: The website would display a summary of the trip, total estimated expenses, remaining budget, and a message showing whether the user was over or under budget.

Conversation 4: Data Abstraction

Topic Discussed:

What information my program would need to store and organize.

Summary of the Project Planning:

The main things my website would need to track include:

- Trip
- Destination
- Budget
- Expenses
- Activities

For each trip, the most important information would be the destination, number of travel days, budget, and estimated expenses.

Other details, such as hotel ratings, flight numbers, weather, and transportation companies, would not be necessary for the basic budgeting calculations.

This helped me understand how to focus on the information that my program actually needs.

Conversation 5: Potential Coding Challenges

Topic Discussed:

What parts of my website might be difficult to implement.

Summary of the Challenges Identified:

Some possible challenges include creating multiple website pages, organizing user information, updating budget calculations, and managing lists of expenses.

Another challenge would be handling unexpected user inputs without causing errors.

The discussion focused on keeping the project simple enough to build with the Python and Drafter concepts learned in class.

Conversation 6: Using AI Responsibly

Topic Discussed:

How I could use ChatGPT while making sure I actually understand my code.

Summary of the Strategy:

I plan to use ChatGPT to help explain programming concepts and error messages instead of copying code without understanding it.

I can ask AI to break down complicated sections into smaller steps, explain how functions and loops work, and help connect the code to what we learned in class.

I will also test the code myself and make sure it works with Drafter before using it in my project.

Conversation 7: Debugging and Resources

Topic Discussed:

What I should do if I get stuck or experience errors while building my website.

Summary of the Debugging Plan:

I would first read the error message carefully and look for common problems such as spelling mistakes, incorrect indentation, missing parentheses, or incorrect variable names.

Then I would test smaller sections of my code to identify where the problem is happening.

If I still could not fix it, I could compare my code with lecture examples, review Drafter documentation, or ask ChatGPT to explain the error.

Finally, I would test my website again to make sure the problem was fixed. I could also ask my professor or TA for additional help.

Overall Reflection

Using ChatGPT helped me organize my project idea and understand which features would be realistic to include in an introductory programming project.

Originally, I wanted to create a website with a detailed travel itinerary and several planning features. After brainstorming, I decided to focus mainly on budgeting, expenses, and a simple activity list.

My final goal is to create a website where users can enter a destination, set a budget, add activities and expenses, and see whether their planned trip is within their budget.

ChatGPT was used as a brainstorming and learning resource throughout the planning process.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.